

JEE MAIN REVISION MASTER: LAWS OF MOTION

2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Laws Of Motion - Core Application	11 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Laws Of Motion - Core Application (High Yield — 11 PYQs)

Core Principle: Key problem pattern in Kcet Laws Of Motion.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string...." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 6m/s^2 . Wha..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A wooden block of mass M lies on a rough floor. Another wooden block of the same mass is hanging from the point O through strings as shown i..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass M is hung with a light inextensible string as shown in figure. Find the tension of the horizontal string.

- (A) $\sqrt{2}Mg$
- (B) $\sqrt{3}Mg$
- (C) Mg
- (D) $3Mg$

Correct Answer: None

Solution Analysis: Let T_1 be the tension in the horizontal string, T_2 be the tension in the slanted string, and T_3 be the tension in the vertical string. The mass M is at rest, so it is in equilibrium. The vertical string supports the weight of the mass. Hence, $T_3 = Mg$. Now consider the knot O . Three tensions act on it: T_1 horizontally, T_2 along the slanted string, and T_3 vertically downward. Since the knot is also in equilibrium, the forces in the vertical direction must balance. $T_2 \sin 45^\circ = T_3 = Mg$. The forces in the horizontal direction must also balance. $T_1 = T_2 \cos 45^\circ = Mg$.